

PLATFORM & DISTRIBUTED SYSTEMS

Distributed Systems

Real-time Streaming

Apache Beam / Dataflow

Pub/SubKubernetes / GKE

IstioGitOps · ArgoCD

Low-latency Services

AI PLATFORM & LLM INFRASTRUCTURE

LLMsLoRA / QLoRALoRAPEFT

UnslothvLLMTransformers

Fine-tuningModel Evaluation

LLM Deployment

ML & DEEP LEARNING

TensorFlowXGBoost

RAPIDS (cuDF/CuPy)

Computer VisionAutoencoders

Distributed Training

Feature Engineering

INFRASTRUCTURE & DELIVERY

Cloud & Orchestration

GCP Vertex AIGKEDataflow

GCSAzure ML Studio

Serving & Packaging

KubernetesHelmDocker

Seldon CoreMLflow

Model Registry

Observability & CI/CD

GrafanaPrometheus

Cloud BuildArgoCD

Google KCC

LANGUAGES & ARCHITECTURE

PythonSQL

Software Architecture

Testing & Debugging

CERTIFICATIONS

Oracle Cloud Infrastructure 2024
Generative AI Certified Professional

Google Cloud

Lead AI and ML engineer and data scientist with **10+ years** of experience, building production AI/ML/LLM systems. Leads end-to-end AI delivery, combining hands-on model development with technical leadership to turn business problems into scalable, high-throughput, low-latency production systems. Experience spans predictive modelling, forecasting, operations research, computer vision, and generative AI across retail, cybersecurity, manufacturing, insurance, and supply-chain optimization. Selected contributions include modelling on **billion-row security datasets**, supply-chain optimization generating **~\$2.5M** in savings, and self-hosted LLMs saving an estimated **~\$300K** annually while improving availability during peak demand.

WORK EXPERIENCE

- Lead Artificial Intelligence Engineer

Best Buy · Bengaluru, India

Jul 2025 - Present

 - Built fraud systems across batch, streaming, and real-time models. Beam/Dataflow feature pipelines handled **40,000 events/sec** at peak; supported a **200 ms scoring SLA** with secure, monitored GKE/Seldon Core inference and load testing.
 - Architected a config-driven Python fine-tuning and automated-evaluation framework (LoRA/QLoRA, Unsloth, PEFT, Accelerate) with vLLM serving; replaced external model APIs, with estimated annual API-cost savings of **~\$300K** and bringing model data under internal governance. Kept model inputs company-managed and reduced external API exposure during peak demand.
 - Lead architecture and production delivery across search, personalization, fraud, and cybersecurity; partner with data scientists to turn validated models into secure, monitored services.
- Lead ML Engineer - ML Engineering & Operations

Wipro · Client: Major US Retail Corporation

Aug 2023 - Jul 2025

 - Led ML/MLOps engineers and Morpheus/RAPIDS model lifecycles on **NVIDIA DGX (8×A100)**, training per-entity autoencoders on **billion-row CloudTrail datasets** with continuous evaluation, debugging, and Grafana alerting. Completed **300+ résumé screens** and **50+ technical interviews** at Wipro.
 - Launched phishing classification at **2,400 emails/sec - 2× prior throughput** and OCR-based abuse-inbox triage that doubled case volume, integrating predictions with Microsoft Sentinel and security-response workflows.
- Senior Data Scientist — Data Science / Azure ML

Accenture

May 2021 - Sep 2023

 - Led cross-functional architecture and launch of an Azure ML quality-prediction product: **\$2.5M annual savings**, inspections **130 → 160/day**, **+17%** defect detection, and **+20%** customer approval.
 - Operationalized data preparation, optimization, retraining, monitoring, and CI/CD; designed OR-Tools services, mentored engineers, and coordinated business, data science, platform, and client stakeholders.

EARLIER EXPERIENCE

Data Scientist

GD Research Centre Pvt Ltd

May 2020 - May 2021

Designed and deployed Python forecasting pipelines across 15+ industry verticals, from heterogeneous data preparation through client-facing results.

Deep Learning Engineer

Tata Consultancy Services

Jul 2019 - Mar 2020

Built and deployed an end-to-end semiconductor ML system: distributed training on **~1 million images in ~3 hours; 90% classification accuracy**. Deployed at Amkor, South Korea, and trained the client team.

Senior Software Engineer - ML

Infosys

May 2016 - Jul 2019

Delivered design, coding, unit/integration testing, and production support; transitioned to Python ML in 2018 and shipped lead-conversion prediction.

Certified Generative AI Leader

Microsoft Certified

Azure AI Engineer Associate

Microsoft Certified

Azure Data Scientist Associate

EDUCATION

B.E. Electrical & Electronics

Andhra University, Visakhapatnam

2012 - 2016 · 80%

GPU & DISTRIBUTED
COMPUTING

CUDA GPU Libraries

GPU-Accelerated Computing

OPTIMISATION & SIMULATION

Google OR-Tools

Vehicle Routing

Mixed-Integer Programming

Constraint Modelling

Operations Research

Supply-Chain Optimization

MODEL LIFECYCLE

Experiment Tracking

Automated Retraining

Vertex AI Pipelines

Azure ML Pipelines

Vertex AI Feature Store

PRODUCTION ENGINEERING

ML System Architecture

Cloud-Native ML Systems

JWT Authentication

Load Testing

Unit / Integration Testing

ENGINEERING LEADERSHIP

Architecture Reviews

Technical Mentoring

Technical Hiring

Team Building

Stakeholder Alignment

Cross-Functional Delivery

PROJECTS

Real-time Streaming & Feature Platform

Best Buy · Aug 2025 - Jan 2026

Large-scale Beam/Dataflow pipelines transform clickstream and transaction events into Vertex AI Feature Store signals. XGBoost models deploy through Seldon Core on GKE with JWT security, load testing, GitOps releases, and production monitoring.

Apache Beam

Vertex AI

40k feature events/sec · 200 ms scoring SLA · Full lifecycle deployment

LLM Fine-tuning, Eval & Serving Platform

Best Buy · Jul 2025 - Present

Config-driven Python platform for LoRA/QLoRA fine-tuning, data preparation, multi-task training, automated evaluation, artifact management, and vLLM serving. Supports Unsloth and PEFT/Transformers with Accelerate; adapters hot-load into a shared GKE deployment.

~\$300K estimated annual API-cost savings

AWS Account Behavioral Fingerprinting

Wipro · Jan 2024 - Aug 2025

Per-entity autoencoders learn normal AWS account, user, service, and machine activity. Morpheus/RAPIDS training and inference run on DGX (8×A100) over billion-row CloudTrail datasets with continuous retraining and Grafana alerts.

RAPIDS / cuDF

Billion-row datasets · Continuous model lifecycle · 8×A100 DGX

Azure ML Product Quality Platform

Accenture / BamaGruppen · May 2021 - Dec 2021

Led an end-to-end ML product from data preparation and model optimization through automated retraining, CI/CD, monitoring, client demonstrations, and ongoing production support on Azure.

Azure ML

Logic Apps

CI/CD

Monitoring

\$2.5M/year · +17% detection · +20% approval · Launched product

ADDITIONAL ENGINEERING PROJECTS

GitOps Delivery Platform

Best Buy · Cloud Build · ArgoCD · Google KCC

Reusable delivery patterns for model services, pipelines, and GCP resources with monitored, zero-downtime releases to GKE.

vLLM Model Chooser

Open source · ylnhari.github.io/vllm-model-chooser

Interactive model and GPU-fit tool covering 115 vLLM-compatible models across VRAM, quantization, and KV-cache constraints.

Rover

Open source · github.com/ylnhari/rover

Remote command executor and project launcher delivered as a dependency-free static binary with real-time event streaming.

ACHIEVEMENTS

15 Awards & Recognitions

Best Buy · Wipro · Accenture · Infosys

Recognised across employers for production impact, reliability, delivery, and engineering contribution.

Talent Quotient 2026

Best Buy

Recognised for cybersecurity and fraud delivery at Best Buy.

Client Recognition

Best Buy client · Wipro engagement

Best Buy client AML Hall of Fame (2024), recognising DGX cybersecurity delivery during the Wipro engagement.

SYSTEM OWNERSHIP & ENGINEERING LEADERSHIP

Architecture & Technical Direction

Architecture & Engineering Standards

Takes undefined platform problems to architecture and execution plans — streaming, serving, evaluation, delivery tooling, observability.

Cross-functional Delivery Distributed, Matrixed Teams

Aligns data science, platform, security, fraud, business, and client teams; coaches engineers and turns ambiguous goals into shipped systems.

Hands-on Through Production Design Through On-call

Writes and reviews production code, debugs live incidents, and owns testing, launch, monitoring and support across cloud and on-prem.